VARDHAMAN
COLLEGE OF ENGINEERING

A Course End Project Report on

Vehicle Rental Management System

A9602- Object oriented programming through java laboratory

Submitted by

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1 Abstract

The Vehicle Rental Management System is a software application developed using Object-Oriented Programming (OOP) concepts to simplify and automate the process of renting vehicles. The system allows users to view available vehicles, select a suitable vehicle, enter customer details, calculate rental charges, and manage rental transactions efficiently. It reduces manual paperwork, minimizes calculation errors, and improves the overall efficiency of vehicle rental services.

The system is designed using fundamental OOP principles such as classes, objects, encapsulation, inheritance, abstraction, and polymorphism. Classes are created to represent vehicles, customers, rental details, and the rental management process. These classes help organize the program into reusable and maintainable components. Different types of vehicles, such as cars, bikes, and vans, can be represented using inheritance.

The proposed system provides a structured approach to vehicle rental operations and can be extended with additional features such as database connectivity, online payments, booking history, and user authentication. This project demonstrates the practical application of Object-Oriented Programming concepts to solve a real-world problem. “‘

2 Introduction

Vehicle rental services provide convenient transportation for people who do not own vehicles or require temporary transportation. Traditional vehicle rental processes often involve manual record maintenance, paperwork, and manual calculation of rental charges. These methods can be time-consuming and may lead to errors in customer records, vehicle availability, and payment calculations.

The Vehicle Rental Management System is designed to automate these activities using Object-Oriented Programming. The system organizes the application into classes and objects, making the program easier to understand, maintain, and extend.

The system maintains information about customers, vehicles, rental duration, and rental charges. Users can view available vehicles, choose a vehicle, enter the rental period, and calculate the total rental cost. The vehicle availability status can also be updated when a vehicle is rented or returned.

OOP principles improve the modularity and reusability of the application. Encapsulation protects data, inheritance supports different vehicle categories, abstraction simplifies system operations, and polymorphism allows common operations to behave differently for different vehicle types.

The main aim of this project is to develop a simple, reliable, and user-friendly vehicle rental application while demonstrating the practical use of Object-Oriented Programming concepts. “

3 Objectives

The main objectives of the Vehicle Rental Management System are:

- To develop a vehicle rental application using Object-Oriented Programming concepts.
- To represent customers, vehicles, and rental transactions using classes and objects.
- To apply encapsulation for protecting customer and vehicle data.
- To implement inheritance for managing different categories of vehicles.
- To demonstrate abstraction and polymorphism in vehicle rental operations.
- To display available vehicles and manage their rental status.
- To calculate rental charges based on the selected vehicle and rental duration.
- To reduce manual work and improve the accuracy of rental operations.
- To develop a modular and maintainable application that can be extended in the future.

4 Methodology

4.1 Block Diagram

The Vehicle Rental Management System consists of the following modules:

- **User Interface:** Allows users to view available vehicles, enter customer details, and select rental options.
- **Vehicle Management Module:** Maintains vehicle information, including vehicle ID, model, type, rental price, and availability status.

- **Customer Management Module:** Stores customer details such as customer ID, name, contact number, and driving licence information.
- **Rental Management Module:** Handles vehicle booking, rental duration, vehicle allocation, and return details.
- **Billing Module:** Calculates the total rental cost based on the rental duration and the vehicle's rental rate.
- **File Handling/Database:** Stores and retrieves vehicle records, customer information, and rental transactions.

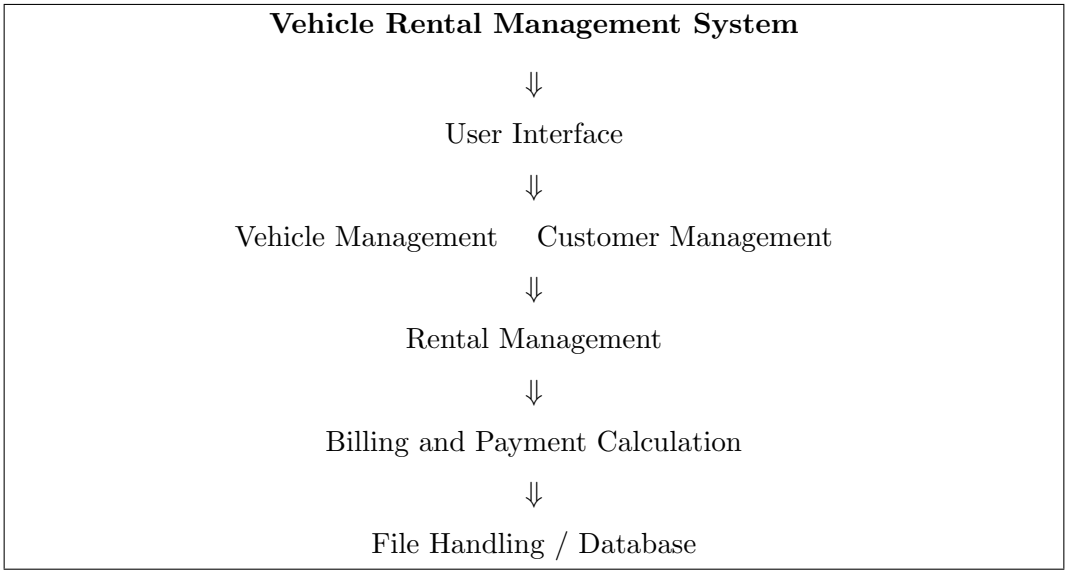


Figure 1: Block Diagram of the Vehicle Rental Management System

The block diagram illustrates the overall structure of the Vehicle Rental Management System. Users interact with the system to view available vehicles, register customer details, book vehicles, and process returns. The management modules handle vehicle availability and booking operations, while the billing module calculates rental charges. The system stores relevant information using files or a database.

4.2 System Design and Components

4.2.1 Components Used

The following components and technologies are required to develop the Vehicle Rental Management System:

- **Programming Language:** C++.
- **Object-Oriented Programming:** Classes, objects, inheritance, encapsulation, and polymorphism.
- **Vehicle Class:** Stores vehicle ID, model, type, rental price, and availability.
- **Customer Class:** Maintains customer information and identification details.
- **Rental Class:** Manages vehicle booking, rental duration, and return operations.

- **Billing Module:** Calculates rental charges based on the selected vehicle and rental period.
- **File Handling:** Saves and retrieves vehicle, customer, and rental records.
- **Compiler/IDE:** Visual Studio Code, Code::Blocks, or another compatible C++ development environment.

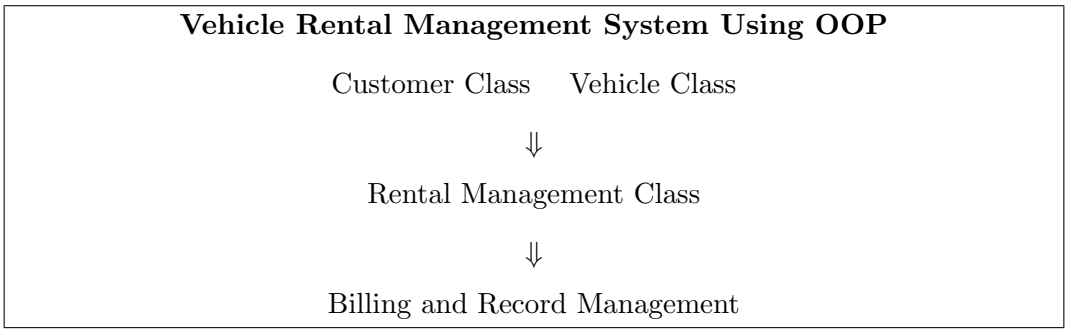


Figure 2: System Design of the Vehicle Rental Management System

The system is developed using C++ and Object-Oriented Programming principles. Customers can select vehicles, provide their details, and rent vehicles for a specified period. The system processes bookings, updates vehicle availability, calculates rental charges, and records vehicle returns. It can be used in car rental agencies, bike rental services, and other vehicle rental businesses.

4.3 Working Principle

The Vehicle Rental Management System operates through the following steps:

1. **System Initialization:** The system loads vehicle information and displays the available operations.
2. **Vehicle Selection:** The user views available vehicles and selects a vehicle based on its model, type, and rental price.
3. **Customer Registration:** The customer enters required details such as name, contact number, and driving licence information.
4. **Vehicle Booking:** The system checks vehicle availability and confirms the booking if the selected vehicle is available.
5. **Rental Charge Calculation:** The system calculates the rental amount based on the daily rental rate and the number of rental days.
6. **Vehicle Return:** When the vehicle is returned, the system records the return details and updates its availability status.
7. **Record Management:** The system stores booking, customer, and billing information for future reference.

The system uses OOP concepts to organize the program into classes and objects. Encapsulation protects data, inheritance supports code reuse, and polymorphism enables related classes to implement common operations in different ways.

4.4 Software Implementation

The Vehicle Rental Management System is implemented using the following software and technologies:

- **C++:** Used to develop the system logic and implement rental operations.
- **Object-Oriented Programming:** Organizes the program using classes, objects, inheritance, encapsulation, and polymorphism.
- **Compiler/IDE:** Used to write, compile, debug, and execute the C++ program.
- **File Handling:** Used to save and retrieve vehicle details, customer information, and rental records when persistent storage is implemented.
- **Database (Optional):** Can be integrated to manage larger amounts of vehicle, customer, and transaction data.
- **Testing and Validation:** Verifies vehicle availability, booking operations, rental charge calculations, and vehicle return processing.

The software implementation provides a structured approach to vehicle rental operations. By applying OOP concepts, the system improves code reusability, maintainability, and organization while reducing manual work in vehicle booking and billing

5 Results and Discussion

The Vehicle Rental System provides an organized and efficient method for managing vehicle details, customer information, rental bookings, and payment records. By using a relational database, the system reduces the need for manual recordkeeping and makes it easier to store, retrieve, and update information. SQL queries allow users to check vehicle availability, retrieve booking details, calculate rental charges, and maintain payment information.

The database uses primary keys and foreign keys to establish relationships between the Customer, Vehicle, Booking, and Payment tables. Integrity constraints help prevent invalid data entries and maintain consistency across related records. JOIN operations allow information from multiple tables to be retrieved in a single query, improving the accessibility of rental information.

The system is suitable for small and medium-sized vehicle rental businesses, including car rental agencies, bike rental services, and travel companies. However, additional mechanisms are required to prevent overlapping bookings, handle concurrent reservations, and secure sensitive customer information. Future enhancements could include online booking, digital payment integration, GPS tracking, automated notifications, and a web-based interface.

The SQL implementation should be tested using the selected DBMS to verify the correctness of database operations and query results.

6 Mapping POs and SDGs

6.1 Program Outcomes (POs) Mapping

6.2 Sustainable Development Goals (SDGs) Mapping

6.3 Sustainable Development Goals (SDGs) Mapping

7 Conclusion

The Vehicle Rental System using a Database Management System provides an organized and reliable approach to managing vehicle rental operations. The system maintains customer details,

Table 1: Mapping of Program Outcomes (POs) to Project Relevance

PO No.	**Program Outcome**	**Relevance**
PO1	Engineering Knowledge	Applied database concepts, relational models, and SQL operations.
PO2	Problem Analysis	Identified challenges in manual vehicle rental record management and booking processes.
PO3	Design/Development of Solutions	Designed a relational database to manage customers, vehicles, bookings, and payments.
PO5	Modern Tool Usage	Utilized MySQL, SQL queries, and database development tools.
PO6	Engineer and Society	Supports efficient vehicle rental operations and organized transportation services.
PO8	Ethics	Highlights the importance of protecting customer information and maintaining accurate records.

vehicle information, booking records, and payment transactions using a relational database. SQL queries, primary keys, foreign keys, and integrity constraints help improve data organization, consistency, and information retrieval.

The project demonstrates the practical application of DBMS concepts in solving real-world problems associated with manual rental management. It can be used as a foundation for developing more advanced vehicle rental applications. Future improvements may include online reservations, secure digital payments, GPS tracking, automated notifications, and mobile application integration to enhance functionality, accessibility, and scalability.

Table 2: Mapping of Sustainable Development Goals (SDGs) to Project Relevance

SDG No.	**Goal**	**Relevance**
SDG 8	Decent Work and Economic Growth	Supports the operational efficiency and digital management of vehicle rental businesses.
SDG 9	Industry, Innovation and Infrastructure	Demonstrates the application of database technology to improve business information systems.
SDG 11	Sustainable Cities and Communities	Can support organized vehicle rental and shared transportation services.
SDG 12	Responsible Consumption and Production	Encourages digital recordkeeping and may reduce paper consumption in rental operations.

Table 3: Mapping of Sustainable Development Goals (SDGs) to Project Relevance

SDG No.	**Goal**	**Relevance**
SDG 9	Industry, Innovation, and Infrastructure	Promotes smart infrastructure in educational institutions.
SDG 11	Sustainable Cities and Communities	Reduces paper wastage by replacing traditional notices.
SDG 12	Responsible Consumption and Production	Encourages digital transformation, reducing environmental impact.

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